Introduction

"Quantum mechanics is the description of the behavior of matter and light in all details and, in particular, of the happenings on an atomic scale" said *R.Feynman* in one of his lectures [1]. In this framework, position is no more a point value unequivocally defined (before measurement), but is instead described by a probability density extending in space (due to wave-particle duality). Consequently, the notion of a trajectory as a continuous path in spacetime is lost in quantum dynamics. We will see further below how to define these trajectories in the case of our study, but one question remains paramount: how can we even define a quantum system?

In the ideal case, systems are treated as *closed*, *i.e.*, isolated from any external environment and interacting with nothing at all. This modelling allows quantum mechanics to describe their dynamics in unitary and deterministic terms (the state at time t is determined solely by knowing the state at time t_0) using the Schrödinger equation. [2]
However in the physical world, systems do "feel" their surroundings, *i.e.*, they interact with their environment. This interaction should be taken into account to obtain a more complete and physically relevant representation; in doing so, we describe the so-called *Open Quantum System* (OQS) theory [3].

Useful theories have been developed through the last decades, allowing quantum description to be more and more complete. One of them is the Quantum Measurement Theory (QMT) [4], where, the action of an observer interacting with a system in order to obtain information defines the concept of measurement. Measuring a system has a tremendous impact on the quantum world, we will take time to explain why further below. However, it is the very role of any experiment to interact with a system in order to measure it. The inquiry of measuring a system without impacting its quantum properties too much leads to the concept of weak measurements.

After studying QMT, one can thus remark that measuring a quantum system alters it. The measurement usually interacts so strongly with the system that it **collapses** into a state — an eigenstate of the observable — that lost interesting quantum properties, *e.g.*, losing entanglement[1]. Or, quantum systems can be in superposition of states, taking *Schrödinger's cat* thought experiment: the cat can be alive and dead at the same time before interaction. Measuring the state of the cat will determine its fate, alive or dead. The state is then "stuck" and cannot be written as a superposition again. The idea of **weak measurements** is to **measure the environment** coupled with the system in order to obtain pieces of information about the system of interest, instead of altering it. This way, information obtained should not alter too much the system due to measurement process. The measurements can even be conducted continuously, maximizing the amount of data the observer can get. These are the global ideas behind **Quantum Continuous Measurements** (QCM) [5].

This work is organized as follows: first section covers an academical progression from quantum closed systems up to the evolution of open quantum systems under continuous photodetection/homodyne detection; following the qubit and harmonic oscillator examples. The second part is a study of a system of interest on which we apply quantum continuous measurement. (Si on a le temps.)

[1] For example, $|\psi\rangle_{AB} = \frac{|00\rangle+|11\rangle}{\sqrt{2}} \xrightarrow{P_0} |00\rangle_{AB} = |0\rangle_A \otimes |0\rangle_B$ a product state (thus non-entangled).

References

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